SIMATS ENGINEERING

ASSESSMENT TOOL 3 – CASE STUDY

Course Name: Internet Programming
Course Code: CSA43

Course Outcome CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery (L2).

Student Name: Sreeshanth Basineni

Register Number: 192372238

Scenario: Web Development Standards Implementation

A start-up developing a modern web application must embed a promotional video, securely submit registration data, use semantic navigation markup, correct XML syntax errors, and communicate through the appropriate web protocol. The following answers select the best option for each requirement and justify the choice.

1. Correct HTML Tag for Video Embedding

Correct option: B. <video src="video.mp4" controls></video>

The HTML <video> element is the standard element for embedding video content in a web page. The src attribute specifies the video file, while the controls attribute displays the browser's playback controls, such as play, pause, and volume. Option A (<media>), option C (<movie>), and option D (<vid>) are not standard HTML elements for video embedding. For better compatibility, a source element can also be used inside video, and fallback text can be included.

```
Example:
<video controls width="640">
  <source src="video.mp4" type="video/mp4">
  Your browser does not support the video element.
</video>
```

2. Appropriate HTTP Method for Form Submission

Correct option: B. POST

POST is the most appropriate choice among the given options for submitting registration data because the form data is sent in the HTTP request body rather than being placed in the URL query string. This helps avoid exposing personal information in browser history and copied URLs. GET is mainly intended for retrieving resources; PUT is commonly used to create or replace a resource at a known target URI; DELETE requests removal of a resource. POST does not itself encrypt data, so the form must use HTTPS, and the server must validate input and handle it securely.

```
Example:
<form action="/register" method="post">
  <label for="username">Username:</label>
  <input id="username" name="username" required>
  <button type="submit">Register</button>
```

```
</form>
```

3. Proper Semantic Element for Navigation

Correct option: B. <nav>

The <nav> element semantically identifies a section containing major navigation links, such as links to the Home, About, Services, and Contact pages. It improves the meaning and structure of the document and helps assistive technologies identify navigation regions. A is a generic container without navigation meaning; <section> groups a thematic section; and <header> represents introductory content or a group of introductory elements. These elements may be useful in a page, but <nav> is the most suitable element for a navigation menu.

```
Example:
<nav aria-label="Main navigation">
  <ul>
    <li><a href="/">Home</a></li>
    <li><a href="/about">About</a></li>
    <li><a href="/contact">Contact</a></li>
  </ul>
</nav>
```

4. Correct XML Syntax

Correct option: B. <note><to>User</to><from>Admin</from></note>

Option B is well-formed XML because every element has a matching closing tag, elements are properly nested, and the root element <note> is closed. Option A is incorrect because the root <note> element is not closed. Option C uses a mismatched closing tag </from> for the opening <to> element and also fails to close the <from> element correctly. Option D is incomplete because the opening <note> tag is missing the closing angle bracket and the element is not properly closed. XML parsers are strict, so malformed XML may cause the configuration file to fail parsing.

```
Correct XML:
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

5. Correct Protocol for Web Communication

Correct option: B. HTTP

HTTP (Hypertext Transfer Protocol) is the application-layer protocol used for communication between web clients, such as browsers, and web servers. It defines requests and responses, including methods, status codes, headers, and message content. FTP is used for file transfer, SMTP is used to send email, and SNMP is used to monitor and manage network devices. For a modern web application, HTTPS should be used in production because it carries HTTP over TLS, protecting data in transit against eavesdropping and tampering.

Typical flow: Browser → HTTPS request → Web server → HTTPS response → Browser renders the content.

Conclusion

The case study demonstrates that following web standards improves compatibility, security, accessibility, and reliability. The correct choices are the <video> element for video, POST for registration submission, <nav> for navigation, option B for well-formed XML, and HTTP for web communication. In deployment, HTTPS should be used to protect information in transit, and HTML/XML should be validated to detect syntax errors early.

Code of Conduct

I, Sreeshanth Basineni (Register Number: 192372238), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sreeshanth Basineni

Date: _____